Devesh Singh

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Professional Experience

Universitätsmedizin Rostock,
Wissenschaftlicher Mitarbeiter / Research Scientist
04/2025 – Present | Rostock
Operationalising the Open Medical Inference (OMI) platform: AI models packaged as Docker-based remote services at partner sites, on interoperable data networks (HL7 FHIR, MII infrastructure).

Benchmarking CT reconstruction (FBP, ASIR, DLIR) against gold-standard MRI with radiologists; Python/MATLAB pipelines for DICOM anonymisation and NIfTI processing.

German Center for Neurodegenerative Diseases (DZNE),
Wissenschaftlicher Mitarbeiter / Research Scientist
09/2022 – 03/2025 | Rostock
Developed explainable deep learning models to help clinicians detect Alzheimer's disease from 3D brain MRI, with a focus on clinical utility and trust.

Combined CNN scoring with a guideline-grounded LLM (RAG) and designed a taxonomy for evaluating hallucinations and output reliability (MIDL 2026).

Data Science For Social Good (DSSGx),
Research Fellow
06/2022 – 08/2022 | Kaiserslautern
Worked with Paraguay's public procurement agency (DNCP) to detect fraud in procurement processes.

SICK Sensor Intelligence, *Master's Thesis*
11/2021 – 04/2022 | Hamburg
Built and tested a synthetic data pipeline with generative models (CycleGAN, CUT) for industrial image translation.

SMA Solar Technology, *Working Student R&D*
03/2021 – 08/2021 | Remote (Germany)
Forecasted inverter failure risk and integrated legacy code using GitFlow in Azure DevOps.

STIHL, *Data Science and Big Data Analytics Intern*
03/2020 – 09/2020 | Waiblingen
Statistical tests on damage correlation with usage, time-series analysis of machine signals, data collection and Power BI dashboards.

Skills

Deep Learning Stack
PyTorch, TensorFlow, Keras

Machine Learning Libraries
Pandas, Numpy, scikit-learn

Coding Languages
Python, R, MATLAB, SQL

Engineering Tools
Docker, Git, Azure DevOps, pytest, Linux, JIRA

Medical Data
DICOM, NIfTI, HL7 FHIR

Database Stack
PostgreSQL, Microsoft SQL, PySpark, Hadoop

Dashboarding Platforms
Power BI, KNIME

Education

Ph.D Student,
University Medicine Rostock and DZNE
09/2022 – present | Rostock, Germany
Thesis submitted June 2026, awaiting defense (Dr. rer. hum.); Explainable AI and Medical Imaging

M.Sc. Data and Knowledge Engineering,
Otto von Guericke University (OvGU)
10/2018 – 06/2022 | Magdeburg, Germany
GPA: 1.3
Machine Learning | Deep Learning | Generative Models | Information Retrieval | Heuristic Optimization | Business Analytics

B.Tech. Information and Communication Technology, *Dhirubhai Ambani Institute of Information and Communication Technology*
08/2013 – 05/2017 | Gandhinagar, India
Software Engineering | Database Management | Computer Networks | OOP Concepts

Publications

Hybrid Multimodal Late Fusion Frameworks for bvFTD Classification in Imbalanced Dementia Datasets, *Frontiers in Aging Neuroscience*
2026

Evidence-Supported Automated Impressions for Alzheimer's Disease Detection from Brain MRI: A Feasibility Study, *Medical Imaging with Deep Learning (MIDL), Short Papers*
2026

An Unsupervised XAI Framework for Dementia Detection with Context Enrichment,
Scientific Reports
2025

SMAS: Structural MRI-Based AD Score using Bayesian VAE,
Computers in Biology and Medicine
2025

Computational Ontology and Visualization Framework for the Visual Comparison of Brain Atrophy Profiles,
BVM Workshop, Springer Vieweg, Wiesbaden
2024
Project GitHub

Contrastive Self-supervised Learning for Neurodegenerative Disorder Classification,
Frontiers in Neuroinformatics
2025

Evaluating the Fidelity of Explanations for Convolutional Neural Networks in Alzheimer's Disease Detection,
Springer Fachmedien Wiesbaden GmbH
2025

Comparison of CNN Architectures for Detecting Alzheimer's Disease using Relevance Maps,
BVM Workshop. Wiesbaden: Springer Fachmedien Wiesbaden
2023
Project GitHub

Languages

English

German
Enrolled for B1.2 level

Other Academic Efforts

Co-editor and reviewer for the Journal of Alzheimer's Disease. Reviewer for Biological Psychiatry, Computers in Biology and Medicine, BMC Neuroscience, Diagnostics (MDPI) and IBRO Neuroscience.

Co-supervised a Master's thesis on large language models and retrieval-augmented generation.

Standardized Test Scores

International English Language Testing System (IELTS)
2017
Score: 8.0

Graduate Record Examination (GRE)
2016
Score: 318

References

Dr. Martin Dyrba, *Junior Group Leader*, DZNE
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